Employee Management System

Java Program Report

1. Introduction

This is a console-based Employee Management System written in Java. It allows the user to create an employee record, display employee details, and increase the employee salary. The program demonstrates classes, objects, methods, loops, conditional statements, Scanner input, and validation.

2. Objectives

- Create and store an employee record.
- Validate the employee name and age.
- Select Programmer, Manager, or Tester as designation.
- Display employee information and salary.
- Increase the employee salary.

3. Description of the Program

Employee class: Stores name, age, designation, and salary. The default salary is ■99,999.

create(): Accepts name, age, and designation. It validates the name, restricts age to 16–66, and lets the user edit the details.

display(): Shows the saved employee details and salary. If no employee exists, it shows an appropriate message.

raiseSalary(): Reads a salary increase, adds it to the current salary, and displays the updated amount.

main(): Shows the menu and repeatedly executes the selected operation until the user chooses Exit.

4. Program Flow

Start → Menu → Create / Display / Raise Salary → Perform Operation → Return to Menu → Exit.

5. Code Screenshot

BankingSystemCode2.java

```java
package BankingSystemProgram;

import java.util.Scanner;

class Employee {
    String name;
    int age;
    String designation;
    double salary = 99999;
}

public class BankingSystemCode2 {
    static Scanner sc = new Scanner(System.in);
    static Employee e = new Employee();

    static void create() {
        while (true) {
            while (true) {
                System.out.print("Enter your name: ");
                e.name = sc.nextLine();
                int spaces = 0;
                for (int i = 0; i < e.name.length(); i++) {
                    if (e.name.charAt(i) == ' ')
                        spaces++;
                }
                if (e.name.isEmpty())
                    System.out.println("Name cannot be empty!");
                else if (spaces > 2)
                    System.out.println("Maximum 2 spaces allowed!");
                else
                    break;
            }

            while (true) {
                System.out.print("Enter your age: ");
                e.age = sc.nextInt();
                sc.nextLine();
                if (e.age >= 16 && e.age <= 66)
                    break;
                System.out.println("Age must be between 16 and 66!");
            }

            while (true) {
                System.out.println("\n1. Programmer");
                System.out.println("2. Manager");
                System.out.println("3. Tester");
                System.out.print("Select designation: ");
                int d = sc.nextInt();
                sc.nextLine();

                if (d == 1) {
                    e.designation = "Programmer";
                    break;
                } else if (d == 2) {
                    e.designation = "Manager";
                    break;
                } else if (d == 3) {
                    e.designation = "Tester";
```

6. Complete Source Code

```
package BankingSystemProgram;

import java.util.Scanner;

class Employee {
    String name;
    int age;
    String designation;
    double salary = 99999;
}

public class BankingSystemCode2 {
    static Scanner sc = new Scanner(System.in);
    static Employee e = new Employee();

    static void create() {
        while (true) {
            while (true) {
                System.out.print("Enter your name: ");
                e.name = sc.nextLine();
                int spaces = 0;
                for (int i = 0; i < e.name.length(); i++) {
                    if (e.name.charAt(i) == ' ')
                        spaces++;
                }
                if (e.name.isEmpty())
                    System.out.println("Name cannot be empty!");
                else if (spaces > 2)
                    System.out.println("Maximum 2 spaces allowed!");
                else
                    break;
            }

            while (true) {
                System.out.print("Enter your age: ");
                e.age = sc.nextInt();
                sc.nextLine();
                if (e.age >= 16 && e.age <= 66)
                    break;
                System.out.println("Age must be between 16 and 66!");
            }

            while (true) {
                System.out.println("\n1. Programmer");
                System.out.println("2. Manager");
                System.out.println("3. Tester");
                System.out.print("Select designation: ");
                int d = sc.nextInt();
                sc.nextLine();

                if (d == 1) {
                    e.designation = "Programmer";
                    break;
                } else if (d == 2) {
                    e.designation = "Manager";
                    break;
                } else if (d == 3) {
                    e.designation = "Tester";
                    break;
                } else {
                    System.out.println("Invalid choice!");
                }
            }

            System.out.println("\n--- Employee Details ---");
            System.out.println("Name: " + e.name);
            System.out.println("Age: " + e.age);
            System.out.println("Designation: " + e.designation);

            System.out.print("Do you want to edit? (yes/no): ");
            String answer = sc.nextLine();
            if (answer.equalsIgnoreCase("no")) {
                System.out.println("Employee saved!");
                break;
            }
        }
    }

    static void display() {
        if (e.name == null) {
            System.out.println("No employee records available.");
        } else {
            System.out.println("\n--- Employee Details ---");
            System.out.println("Name: " + e.name);
            System.out.println("Age: " + e.age);
            System.out.println("Designation: " + e.designation);
            System.out.println("Salary: ■" + e.salary);
        }
    }

    static void raiseSalary() {
        if (e.name == null) {
            System.out.println("No employee records available.");
```

```
            return;
        }
        System.out.println("Current Salary: ■" + e.salary);
        System.out.print("Enter salary raise: ■");
        double raise = sc.nextDouble();
        sc.nextLine();
        e.salary = e.salary + raise;
        System.out.println("Salary updated!");
        System.out.println("New Salary: ■" + e.salary);
    }

    public static void main(String[] args) {
        int choice;
        do {
            System.out.println("\n===== Employee Management =====");
            System.out.println("1. Create");
            System.out.println("2. Display");
            System.out.println("3. Raise Salary");
            System.out.println("4. Exit");
            System.out.print("Enter choice: ");

            choice = sc.nextInt();
            sc.nextLine();

            if (choice == 1)
                create();
            else if (choice == 2)
                display();
            else if (choice == 3)
                raiseSalary();
            else if (choice == 4)
                System.out.println(
                    "Thank you for using Employee Management System!"
                );
            else
                System.out.println("Invalid choice!");

        } while (choice != 4);
    }
}
```

7. Validation

• Name cannot be empty. • Maximum two spaces are allowed. • Age must be between 16 and 66. • Designation must be 1, 2, or 3. • Display and salary raise require an existing employee record.

8. Conclusion

The program is a simple example of Java employee management using object-oriented programming and basic input validation. It can later be extended to support multiple employees, employee IDs, database storage, and advanced salary management.